

The PSA blood test, used to screen for prostate cancer, saves few lives and leads to risky and unnecessary treatments for large numbers of men, two large studies have found.

The findings, the first based on rigorous, randomized studies, confirm some longstanding concerns about the wisdom of widespread prostate cancer screening. Although the studies are continuing, results so far are considered significant and the most definitive to date.

The PSA test, which measures a protein released by prostate cells, does what it is supposed to do — indicates a cancer might be present, leading to biopsies to determine if there is a tumor. But it has been difficult to know whether finding prostate cancer early saves lives. Most of the cancers tend to grow very slowly and are never a threat and, with the faster-growing ones, even early diagnosis might be too late.

The studies — one in Europe and the other in the United States — are “some of the most important studies in the history of men’s health,” said Dr. Otis Brawley, the chief medical officer of the American Cancer Society.

In the European study, 48 men were told they had prostate cancer and needlessly treated for it for every man whose death was prevented within a decade after having had a PSA test.

Dr. Peter B. Bach, a physician and epidemiologist at Memorial Sloan-Kettering Cancer Center, says one way to think of the data is to suppose he has a PSA test today. It leads to a biopsy that reveals he has prostate cancer, and he is treated for it. There is a one in 50 chance that, in 2019 or later, he will be spared death from a cancer that would otherwise have killed him. And there is a 49 in 50 chance that he will have been treated unnecessarily for a cancer that was never a threat to his life.

Prostate cancer treatment can result in impotence and incontinence when surgery is used to destroy the prostate, and, at times, painful defecation or chronic diarrhea when the treatment is radiation.

As soon as the PSA test was introduced in 1987, it became a routine part of preventive health care for many men age 40 and older. Experts debated its value, but their views were largely based on less compelling data that often involved statistical modeling and inferences. Now, with the new data, cancer experts said men should carefully consider the possible risks and benefits of treatment before deciding to be screened. Some may decide not to be screened at all.

For years, the cancer society has urged men to be informed before deciding to have a PSA test. “Now we actually have something to inform them with,” Dr. Brawley said. “We’ve got numbers.”

The publication of data from the two new studies should change the discussion, said Dr. David F. Ransohoff, an internist and cancer epidemiologist at the University of North Carolina. “This is not relying on modeling anymore,” he said. “This is not some abstract, pointy-headed exercise. This is the real world, and this is real data.”

Dr. H. Gilbert Welch, a professor of medicine at Dartmouth who studies cancer screening, also welcomed the new data. “We’ve been waiting years for this,” he said. “It’s a shame we didn’t have it 20 years ago.”

Both reports were published online Wednesday by The New England Journal of Medicine. One involved 182,000 men in seven European countries; the other, by the National Cancer Institute, involved nearly 77,000 men at 10 medical centers in the United States.

In both, participants were randomly assigned to be screened — or not — with the PSA test, whose initials stand for prostate-specific antigen. In each study, the two groups were followed for more than a decade while researchers counted deaths from prostate cancer, asking whether screening made a difference.

The European data involved a consortium of studies with different designs. Taken together, the studies found that screening was associated with a 20 percent relative reduction in the prostate cancer death rate. But the number of lives saved was small — seven fewer prostate cancer deaths for every 10,000 men screened and followed for nine years.

The American study, led by Dr. Gerald L. Andriole of Washington University, had a single design. It found no reduction in deaths from prostate cancer after most of the men had been followed for 10 years. Every man has been followed for at least seven years, said Dr. Barnett Kramer, a study co-author at the National Institutes of Health. By seven years, the death rate was 13 percent lower for the unscreened group.

The European study saw no benefit of screening in the first seven years of follow-up.

Screening is not only an issue in prostate cancer. If the European study is correct, mammography has about the same benefit as the PSA test, said Dr. Michael B. Barry, a prostate cancer researcher at Massachusetts General Hospital who wrote an editorial accompanying the papers. But prostate cancers often are less dangerous than breast cancers, so screening and subsequent therapy can result in more harm. With mammography, about 10 women receive a diagnosis and needless treatment for breast cancer to prevent one death. With both cancers, researchers say they badly need a way to distinguish tumors that would be deadly without treatment from those that would not.

When the American and European studies began, in the early 1990s, PSA testing was well under way in the United States, and many expected that the screening test would make the prostate cancer death rate plummet by 50 percent or more. Dr. Brawley was at the cancer institute then, though not directly involved with its prostate cancer screening study. But he saw the reactions.

Some urologists said the study was unethical, because some people would not be screened, and demanded it be shut down, he said. One group of black urologists encouraged black men not to participate because blacks have a greater risk of prostate cancer and it seemed obvious they should be screened.

Some thought that they would see fewer cancer deaths among screened men as quickly as five years. But it became clear that screening would not have a large, immediate effect — if it did, the studies would have been stopped and victory declared. Cancer researchers began turning to less rigorous sources of data, with some arguing that screening was preventing cancer deaths and others arguing it was not.

In the United States, many men and their doctors have made up their minds — most men over age 50 have already been screened, and each year more than 180,000 receive a diagnosis of prostate cancer. In Europe, said Dr. Fritz H. Schröder of Erasmus University, the lead author of the European study, most men are not screened. “The mentality of Europeans is different,” he said, and screening is not so highly promoted.

Both studies will continue to follow the men. It remains possible that the United States study will eventually find that screening can reduce the prostate cancer death rate, researchers say, or that both studies will conclude that there is no real reduction.

“I certainly think there’s information here that’s food for thought,” Dr. Brawley said.

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